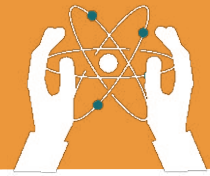


UNIT 2

HOMEOSTASIS

Learning objectives:

- Understand how organisms maintain stable internal conditions.
- Recognize the role of stomata in water regulation and gas exchange in plants.
- Learn how the excretory system helps maintain homeostasis in humans.



2.1 Introduction to Homeostasis

What is Homeostasis?

Homeostasis is the process by which organisms maintain a stable internal environment, essential for the proper functioning of cells, tissues, and organs.

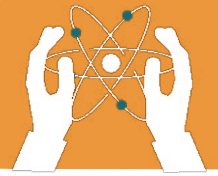
For example, the human body regulates temperature, pH, and water balance for survival.

Why is Homeostasis Important?

Homeostasis is crucial for optimal functioning. If the internal environment deviates too much from the normal range, it can cause diseases or even death.

For example, extreme heat or cold can disrupt organ function, leading to health issues.





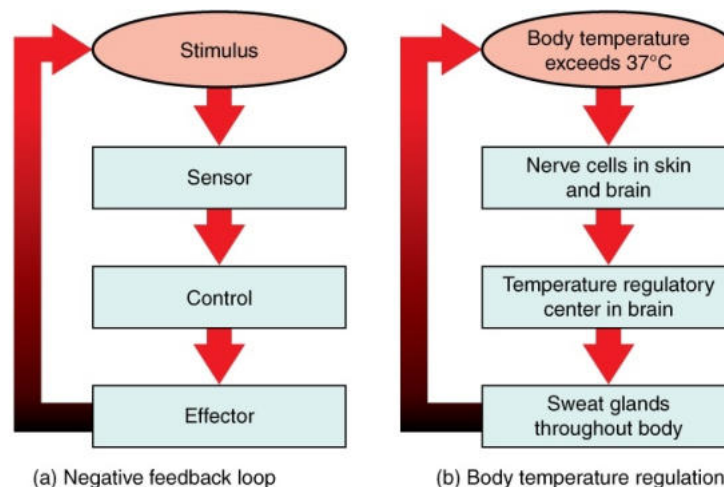
Positive and Negative Feedback in Homeostasis

What is Feedback?

Homeostasis is maintained through feedback mechanisms, which are processes that the body uses to detect changes and restore balance.

Negative Feedback: This mechanism reverses a change to bring the internal environment back to its normal range. For example, when blood sugar levels rise after eating, insulin is released to lower them.

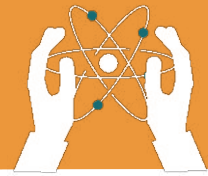
Positive Feedback: This mechanism amplifies a change to drive a process to completion. For instance, upon reaching adulthood, boys start growing moustache and beard.



2.2 Homeostasis in Plants

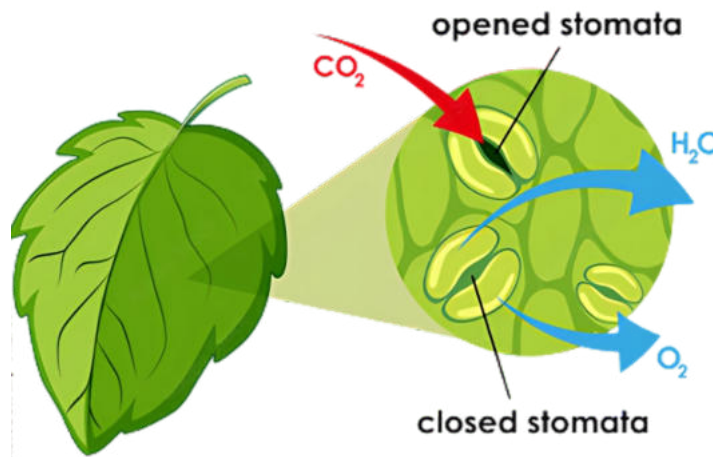
Role of Stomata

In plants, **stomata** are small openings on the leaves that allow gases like oxygen and carbon dioxide to enter and leave the plant. Stomata also play a crucial role in regulating water loss through transpiration. By opening and closing, stomata help maintain the plant's water balance and prevent excessive water loss.



Water Regulation in Plants

Plants regulate water through their roots, leaves, and stems. When water loss is excessive, stomata close to conserve it, while roots absorb water from the soil to maintain hydration and balance.



2.3 Homeostasis in Humans

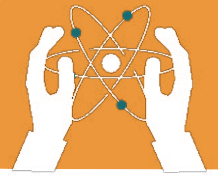
Body Temperature Regulation

Humans maintain a body temperature of around 37°C (98.6°F), which is optimal for enzyme function. When the temperature rises, sweating and dilation of blood vessels help cool the body which is called vasodilation. When it drops, shivering and blood vessel constriction work to warm the body which is called vasoconstriction. These processes ensure temperature stability for proper body function.

Role of Sweat and Shivering

Sweating and shivering are two key responses to regulate body temperature. When it's hot, the sweat glands release sweat, which cools the body as it evaporates. When it's cold, the muscles start to shiver, producing heat to warm the body. Both these processes

HOMEOSTASIS



Shivering



Sweating

2.4 Urinary System in Humans

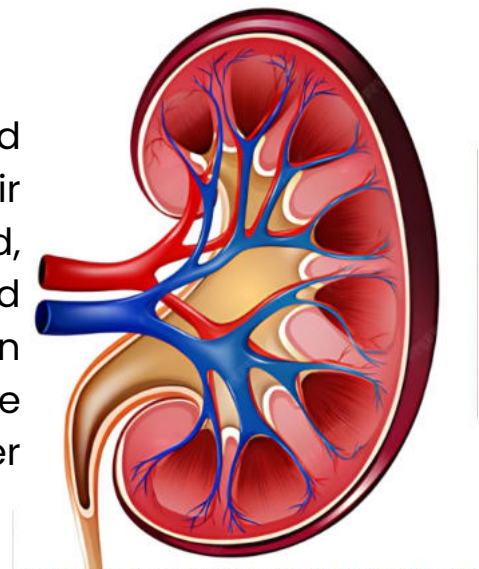
Importance of the Excretory System

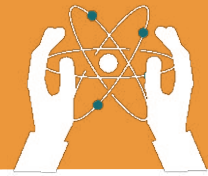
The excretory system in humans is responsible for removing waste products and excess substances from the body, thus maintaining a balanced internal environment.

This system includes the kidneys, ureters, bladder, and urethra. The kidneys filter blood to remove waste, which is then excreted as urine.

Basic Structure and Role of Kidneys

The kidneys are a pair of bean-shaped organs located in the lower back. Their main role is to filter waste from the blood, regulate water and salt balance, and produce urine. The kidneys help maintain homeostasis by ensuring the body has the right balance of water, salts, and other substances.



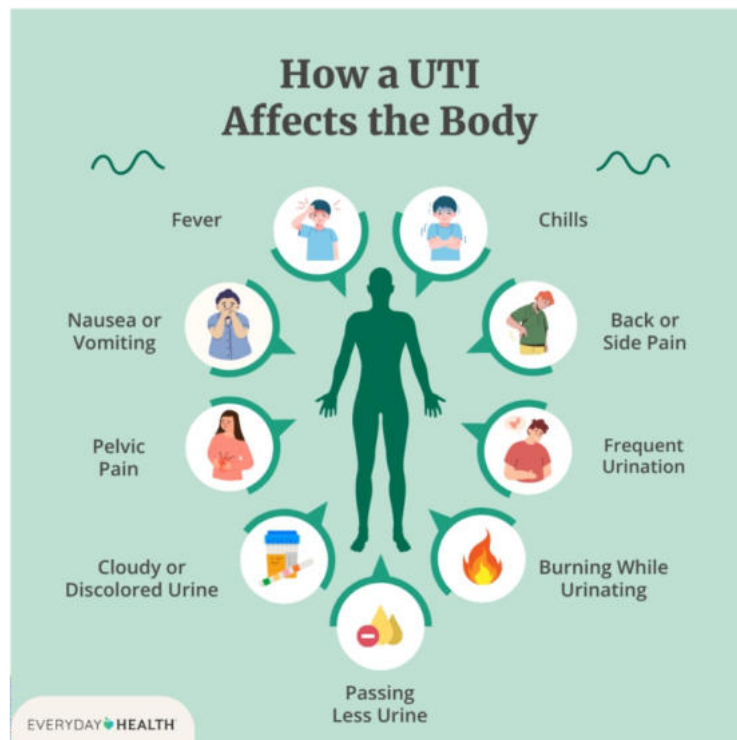


2.5 Disorders of the Excretory System

Common Issues (e.g., Dehydration, Urinary Tract Infections)

Several disorders can affect the excretory system.

- **Dehydration** occurs when the body loses more water than it takes in, leading to an imbalance in body functions.
- **Urinary Tract Infections (UTIs)** are infections in any part of the urinary system, which can cause pain and discomfort during urination.



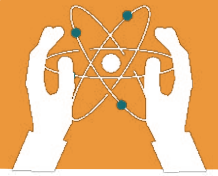
2.6 Healthy Practices for Homeostasis

Tips to Maintain a Healthy Balance (e.g., Balanced Diet, Regular Hydration)

To maintain homeostasis, follow these healthy practices:

- Eat a balanced diet for essential nutrients.
- Stay hydrated by drinking enough water.

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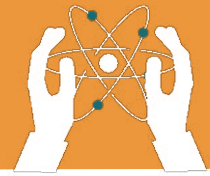


- Exercise regularly to improve circulation and regulate body temperature.
- Get adequate sleep to help the body repair and maintain balance.

EXERCISES:

Fill in the blanks.

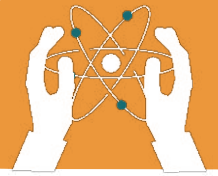
1. Homeostasis helps organisms maintain a stable _____ environment.
2. The stomata in plants regulate _____ loss.
3. The kidneys filter waste products from the _____.
4. _____ is the process by which humans maintain a constant body temperature.
5. Sweating helps cool the body by _____ heat.
6. The kidneys help regulate the balance of _____ in the body.
7. _____ helps prevent dehydration by regulating water levels.
8. The human body regulates temperature through _____ and _____.



HOMEOSTASIS

Tick(✓) for True and cross (x) for False statements.

1. The kidneys help filter waste and regulate water balance. ☐
2. Plants use stomata to absorb carbon dioxide and release oxygen. ☐
3. The excretory system is not important for maintaining homeostasis. ☐
4. Shivering helps cool the human body when it is too hot. ☐
5. Dehydration can be harmful to the body as it disrupts homeostasis. ☐
6. Stomata help prevent water loss from the plant. ☐
7. The liver is responsible for maintaining body temperature. ☐
8. Vasodilation occurs when the body is too cold. ☐



Choose the correct answer.

1. Which of the following is not a role of the kidneys?

- a) Filtering blood b) Regulating water balance
- c) Producing insulin d) Producing urine

2. What process do plants use to regulate water loss?

- a) Photosynthesis b) Transpiration
- c) Respiration d) Germination

3. Which body response occurs when humans are too cold?

- a) Sweating b) Shivering
- c) Vasodilation d) Vasoconstriction

4. The excretory system is responsible for:

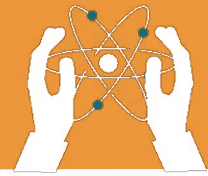
- a) Making energy b) Removing waste from the body
- c) Producing blood d) Digesting food

5. Which of the following can cause dehydration?

- a) Drinking enough water b) High salt intake
- c) Excessive water loss through sweating d) Both b & c

6. The kidneys help in maintaining the balance of _____ in the blood.

- a) Oxygen b) Nitrogen
- c) Water d) Sugar



HOMEOSTASIS

7. When the body is too hot, which of these responses occurs?

- a) Sweating
- b) Shivering
- c) Vasoconstriction
- d) Water retention

8. Which system removes excess water, salts, and waste from the body?

- a) Digestive system
- b) Respiratory system
- c) Excretory system
- d) Nervous system

Answer the Following Questions:

Short-Answer Questions.

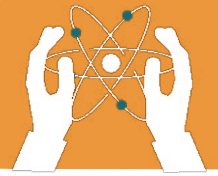
1. What is homeostasis? Why is it important?
2. What is feedback?
3. How do stomata help plants in maintaining homeostasis?
4. What happens when you get too hot or too cold?

Long-Answer Questions.

1. Describe the process of body temperature regulation in humans and the roles of sweating and shivering in maintaining homeostasis.
2. Explain how the excretory system helps in maintaining homeostasis in the human body.

Short Note

1. Write a short note on “Disorders of the excretory system”.



Comparing Breathing Rates



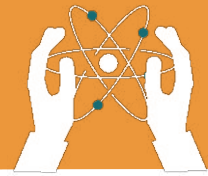
ACTIVITY

Conduct an experiment to observe transpiration in plants. Place a plastic bag around a plant leaf and observe water droplets forming inside the bag over time. Discuss how this demonstrates transpiration.



Critical Thinking

1. How would you explain the importance of hydration in maintaining homeostasis? What could happen if the body doesn't get enough water?
2. What adaptations might an organism develop to maintain homeostasis in an extreme environment, such as a desert or polar region?



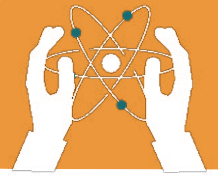
Homeostasis Maintenance Checklist

1. Water Balance

- ☐ Drink 8–10 glasses (2–3 litres) of water daily, adjusting for activity level and climate.
- ☐ Consume water-rich foods (e.g., cucumbers, watermelons).
- ☐ Limit caffeine and alcohol, which can cause dehydration.
- ☐ Monitor urine colour (light yellow indicates proper hydration).

2. Balanced Diet

- ☐ Include a variety of fruits and vegetables in meals (aim for 5 servings daily).
- ☐ Consume lean protein sources (e.g., chicken, fish, beans, or tofu).
- ☐ Choose whole grains (e.g., brown rice, whole wheat bread) for sustained energy.
- ☐ Limit processed foods, added sugars, and unhealthy fats.
- ☐ Ensure adequate intake of essential vitamins and minerals (e.g., calcium, magnesium, potassium).



3. Regular Exercise

- ☐ Engage in at least 30 minutes of moderate physical activity most days of the week.
- ☐ Incorporate strength training exercises 2–3 times per week.
- ☐ Practise flexibility and balance exercises (e.g., yoga or stretching).

4. Quality Sleep

- ☐ Aim for 7–9 hours of uninterrupted sleep each night.
- ☐ Maintain a consistent sleep schedule (same bedtime and wake time).
- ☐ Avoid screens at least 1 hour before bedtime to limit blue light exposure.

Additional Tips

- ☐ Manage stress through mindfulness, meditation, or journaling.
- ☐ Schedule regular health check-ups to monitor vital signs and overall health